

CS 201: Data Structures and Algorithms

Lecture 8: Binary Search Trees and Search Complexities

1. Overview of Tree Structures:

A Binary Tree is a hierarchical structure where each node has at most two children.

Left child and Right child. The top node is called the Root.

A leaf node has zero children.

2. Binary Search Tree (BST) Invariant Property:

For any node X: all keys in X's left subtree are strictly less than X.key.

All keys in X's right subtree are strictly greater than X.key.

3. Search and Insertion Complexity:

Average case search time: $T(n) = O(\log n)$

Worst case search time on degenerate unbalanced tree: $T(n) = O(n)$

Height of balanced tree: $h = \text{floor}(\log_2(n))$

4. Tree Traversals:

Inorder Traversal: Left, Root, Right. Produces sorted sequence for BST.

Preorder Traversal: Root, Left, Right. Used to serialize or clone tree.

Postorder Traversal: Left, Right, Root. Used for deleting tree bottom-up.

5. Exam Key Takeaways:

Always verify BST property holds for all descendants, not just immediate children.